
Examiners' commentaries 2023 (October)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2022–23. The format and structure of the examination may change in future years, and any such changes will be publicised on the virtual learning environment (VLE).

Information about the subject guide and the Essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the subject guide (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list and/or subject guide refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section.

General remarks

Learning outcomes

At the end of this half course and having completed the essential reading and activities you should be able to:

- explain the principles of data reduction
- judge the quality of estimators
- choose appropriate methods of statistical inference to tackle real problems.

Key steps to improvement

Candidates should pay close attention to the essential topics of sufficient statistics, point estimation, interval estimation and hypothesis testing. Being able to state and prove key theorems and lemmas simply requires close study of these in the subject guide. The point estimation techniques of method of moments estimation and maximum likelihood estimation are standard (and are even introduced in **ST104B Statistics 2**), with questions typically varying in terms of the probability distribution from which the random sample is drawn (with many common distributions covered in **ST104B Statistics 2** and **ST2133 Advanced statistics: distribution theory**). Pivotal functions feature heavily in Chapter 4 (Interval estimation), and likelihood ratio tests similarly feature heavily in Chapter 5 (Hypothesis testing).

An examination paper attempts to cover as much of the syllabus as possible, but inevitably some parts are left out. As the included parts will vary from year to year it is not enough to practise only on the past papers. Candidates should practise on all the examples, activities and sample

examination questions in the subject guide in order to understand the relevant theory and adequately prepare for the examination. Afterwards you may try one or two past papers for further practice.

The examination paper for this course involves four questions and each of the questions contains several parts which contribute towards the final grade with the different numbers of marks reflecting their difficulty. The difficulty of each part is usually assessed after taking into account the level of thinking related to statistical inference and the extent of algebraic computations required. Take 5–10 minutes to read the whole paper and try to identify the difficulties of each question in comparison to your ability. You will find that some questions could be much easier for you than others.

If you are unable to solve an exercise due to algebraic computations or lack of knowledge of the material of **ST2133 Advanced statistics: distribution theory**, do not abandon an exercise. Write clearly the procedure which should be followed for the relevant statistical inference task(s). Remember that the primary aspect you are being examined on is the ability to *understand* the statistical methodology.

A good knowledge of basic calculus, such as the ability to compute integrals, sums and derivatives, will give you invaluable help for completing the examination paper. Also, the main concepts of **ST2133 Advanced statistics: distribution theory** (random variables and their distribution, expectation, independence, conditional expectation etc.) should be well-understood before attempting this half course. If you are not comfortable with any of these, go back and revise. Memorising basic properties such as the moments and probability mass/density functions of some standard distributions (such as the binomial, Poisson, uniform, normal and gamma) is also a good idea.

When you are trying to prove an equality or inequality be careful of what is on the left-hand and right-hand sides. Justify the steps in the calculations as much as possible. If you cannot get a question right, do not just write in the correct answer without any further explanation. If you just write on the script what you think went wrong you will get more marks. Also, try to avoid writing more than you are asked to write. For example, when asked to *state* a theorem, do not give the proof as well as this simply wastes time.

Be clear in your answers. When giving the probability mass/density function of a random variable you should also state its sample space (also known as its support). When you are deriving an estimator or a sufficient statistic also state the parameter to which it corresponds. Sometimes it is a good idea to underline your final result (if applicable).

Make sure that you are able to provide the definitions of basic concepts such as estimators vs. estimates, confidence intervals and hypothesis tests. Remember that hypotheses are statements about population parameters which are estimated using functions of the data (that is, *statistics*). In the relevant calculations you should always be able to distinguish a statistic from a parameter and be clear on what they represent, respectively.

Practise operations involving products and sums. A substantial number of candidates fail to write them down correctly. Quite often they omit or write the indices incorrectly which inevitably leads to errors. A typical example of this is the following:

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population with probability density function $f_{X_i}(x_i; \theta)$, often written for simplicity as $f_X(x; \theta)$. An observed sample is $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. The fact that this sample is a random sample indicates that we can write the joint probability density function of $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta).$$

Make sure that you write the above and *not* the following:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x; \theta)$$

which can lead to the *incorrect* expression:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = (f_X(x; \theta))^n.$$

Note that if you were asked to provide the log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$ the calculations would be formed of *sums* rather than *products* (refresh your memory on the properties of logarithmic functions). The correct expression would be:

$$\begin{aligned} l_{\mathbf{X}}(\theta; \mathbf{x}) &= \log(f_{\mathbf{X}}(\theta; \mathbf{x})) \\ &= \log \left(\prod_{i=1}^n f_{X_i}(x_i; \theta) \right) \\ &= \sum_{i=1}^n \log f_{X_i}(x_i; \theta). \end{aligned}$$

Examination revision strategy

Many candidates are disappointed to find that their examination performance is poorer than they expected. This may be due to a number of reasons, but one particular failing is ‘**question spotting**’, that is, confining your examination preparation to a few questions and/or topics which have come up in past papers for the course. This can have serious consequences.

We recognise that candidates might not cover all topics in the syllabus in the same depth, but you need to be aware that examiners are free to set questions on **any aspect** of the syllabus. This means that you need to study enough of the syllabus to enable you to answer the required number of examination questions.

The syllabus can be found in the Course information sheet available on the VLE. You should read the syllabus carefully and ensure that you cover sufficient material in preparation for the examination. Examiners will vary the topics and questions from year to year and may well set questions that have not appeared in past papers. Examination papers may legitimately include questions on any topic in the syllabus. So, although past papers can be helpful during your revision, you cannot assume that topics or specific questions that have come up in past examinations will occur again.

If you rely on a question-spotting strategy, it is likely you will find yourself in difficulties when you sit the examination. We strongly advise you not to adopt this strategy.

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Comments on specific questions

Candidates should answer all **FOUR** questions: **Question 1** of Section A (40 marks) and all **THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

Section A

Answer Question 1 from this section.

Question 1

- (a) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a $\text{Gamma}(\alpha, \beta)$ distribution, with probability density function:

$$f_{\mathbf{X}}(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\beta}\right)$$

for $x \geq 0$, and 0 otherwise, where $\alpha > 0$ and $\beta > 0$ are parameters.

- i. Find a 1-dimensional sufficient statistic for α , when β is known. (4 marks)
- ii. Find a 1-dimensional sufficient statistic for β , when α is known. (3 marks)
- iii. Find a 2-dimensional sufficient statistic for α and β . (3 marks)

Reading for this question

Section 2.8 of the subject guide covers sufficient statistics and the factorisation theorem.

Approaching the question

Due to independence, the joint probability density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \alpha, \beta) = \prod_{i=1}^n f_X(x_i; \alpha, \beta) = \left(\frac{1}{\Gamma(\alpha)}\right)^n e^{-\sum_i x_i/\beta} \left(\prod_{i=1}^n x_i\right)^{\alpha-1} \beta^{-n\alpha}.$$

i. If β is known, then we may set:

$$g\left(\prod_{i=1}^n x_i, \alpha\right) = \left(\frac{1}{\Gamma(\alpha)}\right)^n \left(\prod_{i=1}^n x_i\right)^{\alpha-1} \beta^{-n\alpha} \quad \text{and} \quad h(\mathbf{x}) = e^{-\sum_i x_i/\beta}.$$

Since $f_{\mathbf{X}}(\mathbf{x}; \alpha, \beta) = g\left(\prod_{i=1}^n x_i, \alpha\right) h(\mathbf{x})$, the factorisation theorem implies that the statistic $\prod_{i=1}^n X_i$ is sufficient for α .

ii. If α is known, then we may set:

$$g\left(\sum_{i=1}^n x_i, \beta\right) = e^{-\sum_i x_i/\beta} \beta^{-n\alpha} \quad \text{and} \quad h(\mathbf{x}) = \left(\frac{1}{\Gamma(\alpha)}\right)^n \left(\prod_{i=1}^n x_i\right)^{\alpha-1}.$$

Since $f_{\mathbf{X}}(\mathbf{x}; \alpha, \beta) = g\left(\sum_{i=1}^n x_i, \beta\right) h(\mathbf{x})$, the factorisation theorem implies that the statistic $\sum_{i=1}^n X_i$ is sufficient for β .

iii. If neither α nor β are known, we may set:

$$g\left(\prod_{i=1}^n x_i, \sum_{i=1}^n x_i, \alpha, \beta\right) = f_{\mathbf{X}}(\mathbf{x}; \alpha, \beta) \quad \text{and} \quad h(\mathbf{x}) = 1.$$

Since $f_{\mathbf{X}}(\mathbf{x}; \alpha, \beta) = g\left(\prod_{i=1}^n x_i, \sum_{i=1}^n x_i, \alpha, \beta\right) h(\mathbf{x})$, the factorisation theorem implies that the statistic $\left(\prod_{i=1}^n X_i, \sum_{i=1}^n X_i\right)$ is sufficient for (α, β) .

(b) State and prove the Rao–Blackwell theorem.

(10 marks)

Reading for this question

Section 3.5 of the subject guide presents the Rao–Blackwell theorem.

Approaching the question

Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$. Let $\mathbf{Y}(\mathbf{X})$ be a statistic of the same dimension as $\boldsymbol{\theta}$, and let $\mathbf{S}(\mathbf{X})$ be a sufficient statistic for $\boldsymbol{\theta}$. If we define $\mathbf{T}(\mathbf{X}) = E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X}))$, then $\mathbf{T}(\mathbf{X})$ is a statistic and:

$$\text{MSE}_{\boldsymbol{\theta}}(\mathbf{T}(\mathbf{X})) \leq \text{MSE}_{\boldsymbol{\theta}}(\mathbf{Y}(\mathbf{X})).$$

Proof: $\mathbf{T}(\mathbf{X})$ is a sufficient statistic due to the sufficiency of $\mathbf{S}(\mathbf{X})$. For the mean squared error result, we begin by considering the variance of $\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}$, conditional on $\mathbf{S}(\mathbf{X})$, i.e. we consider:

$$\text{Var}((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})) = E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2 | \mathbf{S}(\mathbf{X})) - (E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})))^2.$$

Since $\text{Var}((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})) \geq 0$ (it is a variance, hence is non-negative), we have:

$$E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2 | \mathbf{S}(\mathbf{X})) \geq (E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})))^2.$$

We can now prove the result directly:

$$\begin{aligned}
 \text{MSE}_\theta(\mathbf{Y}(\mathbf{X})) &= E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2) && \text{(by definition)} \\
 &= E(E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2 | \mathbf{S}(\mathbf{X}))) && \text{(law of iterated expectations)} \\
 &\geq E(E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X}))^2) \\
 &= E((E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X})) - \boldsymbol{\theta})^2) && \text{(properties of expectation)} \\
 &= E((\mathbf{T}(\mathbf{X}) - \boldsymbol{\theta})^2) \\
 &= \text{MSE}_\theta(\mathbf{T}(\mathbf{X})).
 \end{aligned}$$

- (c) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from the following Pareto distribution:

$$f_X(x; \alpha) = \frac{\alpha}{(1+x)^{\alpha+1}}$$

for $0 < x < \infty$, and 0 otherwise, where $\alpha > 0$ is a parameter.

Derive the probability density function of the sample minimum, i.e. $X_{(1)}$.

(10 marks)

Reading for this question

Section 3.7 of the subject guide presents sample quantiles and order statistics, including the sample minimum.

Approaching the question

In general, the probability density function of the sample minimum is:

$$nf_X(x)(1 - F_X(x))^{n-1}.$$

We have that:

$$F_X(x) = \int_0^x \frac{\alpha}{(1+t)^{\alpha+1}} dt = 1 - (1+x)^{-\alpha} \quad \text{and} \quad 1 - F_X(x) = (1+x)^{-\alpha}$$

hence:

$$\begin{aligned}
 f_{X_{(1)}}(x) &= nf_X(x)(1 - F_X(x))^{n-1} \\
 &= n \frac{\alpha}{(1+x)^{\alpha+1}} ((1+x)^{-\alpha})^{n-1} \\
 &= \frac{n\alpha}{(1+x)^{n\alpha+1}}
 \end{aligned}$$

and so $X_{(1)} \sim \text{Pareto}(n\alpha)$.

- (d) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the *negative binomial* distribution with probability function:

$$p_X(x; r, \pi) = \begin{cases} \binom{x-1}{r-1} \pi^r (1-\pi)^{x-r} & \text{for } x = r, r+1, r+2, \dots \\ 0 & \text{otherwise} \end{cases}$$

which is the distribution of the number of Bernoulli trials for r successes to occur. The parameter π is the probability of success at each trial. You may use the fact that for this distribution $E(X) = r/\pi$ and $\text{Var}(X) = r(1-\pi)/\pi^2$.

Derive the method of moments estimators of r and π .

(10 marks)

Reading for this question

Section 3.6.5 of the subject guide presents method of moments estimation.

Approaching the question

Since there are two parameters we require the first two population moments, i.e. $E(X) = r/\pi$ and:

$$E(X^2) = \text{Var}(X) + (E(X))^2 = \frac{r(1-\pi)}{\pi^2} + \frac{r^2}{\pi^2} = \frac{r(1-\pi) + r^2}{\pi^2}.$$

The first moment equation then gives:

$$\frac{\hat{r}}{\hat{\pi}} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \quad \Rightarrow \quad \hat{r} = \hat{\pi} \bar{X}.$$

Using this result, the second moment equation then gives:

$$\frac{\hat{r}(1-\hat{\pi}) + \hat{r}^2}{\hat{\pi}^2} = \frac{\hat{\pi} \bar{X}(1-\hat{\pi}) + (\hat{\pi} \bar{X})^2}{\hat{\pi}^2} = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

Equivalently:

$$\hat{\pi} \bar{X} - \hat{\pi}^2 \bar{X} + \hat{\pi}^2 \bar{X}^2 - \frac{\hat{\pi}^2}{n} \sum_{i=1}^n X_i^2 = 0.$$

Solving for $\hat{\pi}$ gives:

$$\hat{\pi} = \frac{\bar{X}}{\bar{X} + \sum_{i=1}^n X_i^2/n - \bar{X}^2}.$$

Finally:

$$\hat{r} = \frac{\bar{X}^2}{\bar{X} + \sum_{i=1}^n X_i^2/n - \bar{X}^2}.$$

Section B

Answer all three questions from this section.

Question 2

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the Uniform $[\theta_1, \theta_2]$ distribution, i.e. the continuous uniform distribution over the interval $[\theta_1, \theta_2]$, with $\theta_1 < \theta_2$.

- (a) Derive the first and second population moments of the probability distribution Uniform $[\theta_1, \theta_2]$.
(6 marks)
- (b) Suppose that $\theta_1 = \theta_2 - 3$. Find the method of moments estimator of θ_2 .
(4 marks)
- (c) Find the method of moments estimator of θ_2 when $\theta_1 = -\theta_2$.
(6 marks)
- (d) Derive the equations to be solved (but do not solve them) in order to obtain the method of moments estimators of θ_1 and θ_2 for any case such that $\theta_1 < \theta_2$.
(4 marks)

Reading for this question

Section 3.6.5 of the subject guide presents method of moments estimation.

Approaching the question

(a) The probability density function is:

$$f_X(x; \theta_1, \theta_2) = \begin{cases} 1/(\theta_2 - \theta_1) & \text{for } \theta_1 \leq x \leq \theta_2 \\ 0 & \text{otherwise.} \end{cases}$$

We have:

$$E(X_i) = \int_{\theta_1}^{\theta_2} \frac{x}{\theta_2 - \theta_1} dx = \left[\frac{x^2}{2(\theta_2 - \theta_1)} \right]_{\theta_1}^{\theta_2} = \frac{\theta_2 + \theta_1}{2}.$$

Also:

$$E(X_i^2) = \int_{\theta_1}^{\theta_2} \frac{x^2}{\theta_2 - \theta_1} dx = \left[\frac{x^3}{3(\theta_2 - \theta_1)} \right]_{\theta_1}^{\theta_2} = \frac{\theta_2^3 - \theta_1^3}{3(\theta_2 - \theta_1)} = \frac{\theta_2^2 + \theta_1^2 + \theta_1\theta_2}{3}.$$

(b) When $\theta_1 = \theta_2 - 3$, $E(X_i) = \theta_2 - 1.5$. Hence the method of moments estimator of θ_2 solves:

$$\hat{\theta}_2 - 1.5 = \bar{X} \Rightarrow \hat{\theta}_2 = \bar{X} + 1.5.$$

(c) When $\theta_1 = -\theta_2$, $E(X_i) = 0$, which does not depend on θ_2 . The second moment under this constraint is $E(X^2) = \theta_2^2/3$. Hence the method of moments estimator of θ_2 solves:

$$\frac{\hat{\theta}_2^2}{3} = \frac{1}{n} \sum_{i=1}^n X_i^2 \Rightarrow \hat{\theta}_2 = \left(\frac{3}{n} \sum_{i=1}^n X_i^2 \right)^{1/2}.$$

(d) The method of moments estimators of θ_1 and θ_2 may be defined as the solutions to the equations:

$$\bar{X} = \frac{\hat{\theta}_2 + \hat{\theta}_1}{2} \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n X_i^2 = \frac{\hat{\theta}_2^2 + \hat{\theta}_1^2 + \hat{\theta}_1 \hat{\theta}_2}{3}.$$

Question 3

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $N(0, \theta)$ distribution, with probability density function:

$$f_X(x; \theta) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{x^2}{2\theta}\right) \quad \text{for } -\infty < x < \infty$$

where $\theta > 0$.

(a) Derive the log-likelihood function and find a sufficient statistic for θ .

(3 marks)

(b) Derive the score function and find the Fisher information for θ .

(5 marks)

(c) Derive the maximum likelihood estimator of θ and check whether it is the minimum variance unbiased estimator of θ .

(7 marks)

- (d) Find a pivotal function for θ and construct a $100(1 - \alpha)\%$ confidence interval for θ . Use the exact distribution of your pivotal function.

(5 marks)

Reading for this question

Section 2.10 of the subject guide explains how to derive the log-likelihood and score functions. Maximum likelihood estimation is covered in Section 3.9, while Section 4.6 covers finding interval estimators from pivotal functions.

Approaching the question

- (a) Due to independence, the joint probability density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x_i; \theta) = \prod_{i=1}^n (2\pi\theta)^{-1/2} \exp\left(-\frac{x_i^2}{2\theta}\right) = (2\pi\theta)^{-n/2} \exp\left(-\frac{1}{2\theta} \sum_{i=1}^n x_i^2\right).$$

If we set:

$$g\left(\sum_{i=1}^n x_i^2, \theta\right) = f_{\mathbf{X}}(\mathbf{x}; \theta) \quad \text{and} \quad h(\mathbf{x}) = 1$$

we can write $f_{\mathbf{X}}(\mathbf{x}; \theta) = g\left(\sum_{i=1}^n x_i^2, \theta\right) h(\mathbf{x})$. According to the factorisation theorem, this implies the statistic $\sum_{i=1}^n X_i^2$ is sufficient for θ .

The log-likelihood function is:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \log L_{\mathbf{X}}(\theta; \mathbf{x}) = \log f_{\mathbf{X}}(\mathbf{x}; \theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\theta) - \frac{1}{2\theta} \sum_{i=1}^n x_i^2.$$

- (b) The score function for θ is:

$$s_{\mathbf{X}}(\theta; \mathbf{X}) = \frac{\partial}{\partial \theta} l_{\mathbf{X}}(\theta; \mathbf{x}) = -\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n x_i^2.$$

The second derivative of the log-likelihood function with respect to θ is:

$$\frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{n}{2\theta^2} - \frac{1}{\theta^3} \sum_{i=1}^n x_i^2.$$

Hence the Fisher information is equal to:

$$I_{\mathbf{X}}(\theta) = -E\left(\frac{n}{2\theta^2} - \frac{1}{\theta^3} \sum_{i=1}^n X_i^2\right) = -\frac{n}{2\theta^2} + \frac{\sum_{i=1}^n E(X_i^2)}{\theta^3} = -\frac{n}{2\theta^2} + \frac{n\text{Var}(X_i)}{\theta^3} = -\frac{n}{2\theta^2} + \frac{n}{\theta^2} = \frac{n}{2\theta^2}.$$

- (c) Solving $s_{\mathbf{X}}(\hat{\theta}; \mathbf{X}) = 0$ yields a potential maximum likelihood estimator (MLE). We have:

$$\frac{1}{2\hat{\theta}^2} \sum_{i=1}^n x_i^2 = \frac{n}{2\hat{\theta}} \quad \text{or else} \quad \hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

The second derivative of the log-likelihood function with respect to θ is:

$$\frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{n}{2\theta^2} - \frac{1}{\theta^3} \sum_{i=1}^n x_i^2.$$

If we evaluate the above function at $\theta = \sum_{i=1}^n x_i^2/n$, we get:

$$\left. \frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{x}) \right|_{\theta = \sum_{i=1}^n x_i^2/n} = \frac{n^3}{2 \left(\sum_{i=1}^n x_i^2 \right)^2} - \frac{n^3 \sum_{i=1}^n x_i^2}{\left(\sum_{i=1}^n x_i^2 \right)^3} = -\frac{n^3}{2 \left(\sum_{i=1}^n x_i^2 \right)^2}.$$

As the above only takes negative values we conclude that the MLE of θ is $\hat{\theta} = \sum_{i=1}^n X_i^2/n$.

We can show that the MLE is the minimum variance unbiased estimator (MVUE) in two ways. However, first we have to check whether it is an unbiased estimator of θ . Indeed, since $E(X_i) = 0$ it is because:

$$E \left(\frac{\sum_{i=1}^n X_i^2}{n} \right) = \frac{n E(X_i^2)}{n} = \text{Var}(X_i^2) = \theta.$$

Method 1: The score function may be written as:

$$s_{\mathbf{X}}(\theta; \mathbf{X}) = \frac{n}{2\theta^2} \left(\frac{\sum_{i=1}^n X_i^2}{n} - \theta \right)$$

where $\sum_{i=1}^n X_i^2/n$ is an unbiased estimator of θ and $n/(2\theta^2)$ is only a function of parameters.

The Cramér–Rao lower bound attainment theorem suggests that the MLE attains the Cramér–Rao lower bound and therefore is the MVUE.

Method 2: Since the Fisher information is equal to $n/(2\theta^2)$, the Cramér–Rao lower bound of an unbiased estimator of θ is $2\theta^2/n$.

Since $X_i^2/\theta \sim \chi_1^2$, we have that $\text{Var}(X_i^2/\theta) = 2$, or else $\text{Var}(X_i^2) = 2\theta^2$. Hence:

$$\text{Var}(\hat{\theta}) = \text{Var} \left(\frac{\sum_{i=1}^n X_i^2}{n} \right) = \frac{n \text{Var}(X_i^2)}{n^2} = \frac{2\theta^2}{n}.$$

Since the MLE is unbiased and its variance attains the Cramér–Rao lower bound, it is also the MVUE.

(d) Since each:

$$\frac{X_i - \mu}{\sqrt{\theta}} \sim N(0, 1)$$

and the X_i s are independent, we have:

$$\frac{\sum_{i=1}^n X_i^2}{\theta} \sim \chi_n^2.$$

The function $\sum_{i=1}^n X_i^2/\theta$ is a pivotal function for θ .

Let $\chi_{\alpha, n}^2$ denote the top 100α th percentile of a χ_n^2 distribution. As $\sum_{i=1}^n X_i^2/\theta \sim \chi_n^2$,

$$1 - \alpha = P \left(\chi_{1-\alpha/2, n}^2 \leq \frac{\sum_{i=1}^n X_i^2}{\theta} \leq \chi_{\alpha/2, n}^2 \right) = P \left(\frac{\sum_{i=1}^n X_i^2}{\chi_{\alpha/2, n}^2} \leq \theta \leq \frac{\sum_{i=1}^n X_i^2}{\chi_{1-\alpha/2, n}^2} \right).$$

Hence an exact $100(1 - \alpha)\%$ confidence interval for θ is:

$$\left[\frac{\sum_{i=1}^n X_i^2}{\chi_{\alpha/2, n}^2}, \frac{\sum_{i=1}^n X_i^2}{\chi_{1-\alpha/2, n}^2} \right].$$

Question 4

- (a) Explain the rationale behind the likelihood ratio test and why the likelihood ratio test statistic takes values larger than one. Describe how this test may be used in the presence of nuisance parameters.

(8 marks)

Reading for this question

Section 5.6 of the subject guide presents likelihood ratio tests.

Approaching the question

The likelihood ratio test statistic is connected with maximum likelihood estimation. Consider an observed sample $\mathbf{x}$. Hence the observed value of the likelihood ratio test statistic is:

$$LR(\mathbf{x}) = \frac{\sup_{\theta \in \Theta} L_{\mathbf{x}}(\theta; \mathbf{x})}{\sup_{\theta \in \Theta_0} L_{\mathbf{x}}(\theta; \mathbf{x})}.$$

The numerator is the *global maximum* of the likelihood function, attained at the maximum likelihood estimate $\hat{\theta}$. The denominator is the maximum of the likelihood function *under the constraint* that $\theta \in \Theta_0$. Let $\hat{\theta}_0$ denote the maximum value of this constrained likelihood function. Hence the observed value of the likelihood ratio test statistic can be expressed as:

$$LR(\mathbf{x}) = \frac{L_{\mathbf{x}}(\hat{\theta}; \mathbf{x})}{L_{\mathbf{x}}(\hat{\theta}_0; \mathbf{x})}.$$

This value must be ≥ 1 (since the constrained parameter estimate, $\hat{\theta}_0$, cannot be better than the unconstrained estimate, $\hat{\theta}$, in the sense that the associated likelihood value cannot be larger), although rejection of H_0 only occurs if the value is significantly greater than 1, i.e. when $\hat{\theta}_0$ is significantly worse than $\hat{\theta}$.

Now suppose that θ can be split into two groups of parameters: ψ being the main parameters of interest, and λ being the parameters which are of little interest – known as **nuisance parameters**.

Using this specification, we can think of H_0 as imposing a constraint on the model being fitted. The constraint is that ψ must take the values specified by ψ_0 . Hence the likelihood ratio test statistic compares two models, each with a relative advantage.

- The **unconstrained model** will provide a better fit since the associated likelihood will be larger. We denote the unconstrained maximum likelihood estimate by:

$$\hat{\theta} = (\hat{\psi}, \hat{\lambda}).$$

- The **constrained model** has fewer parameters requiring estimation. If the dimension of ψ is k , then the values of the k parameters contained in ψ are fixed, such that we only need to estimate the nuisance parameters which are in λ , where $\dim(\lambda) < \dim(\theta)$. Under the constraint that $\psi = \psi_0$, the maximum likelihood estimate of λ is generated by maximising:

$$L_{\mathbf{x}}(\psi_0, \lambda; \mathbf{x})$$

with respect to λ . We denote the resulting estimate as $\hat{\lambda}_0$. Hence the constrained maximum likelihood estimate of θ is:

$$\hat{\theta}_0 = (\psi_0, \hat{\lambda}_0).$$

When testing:

$$H_0 : \psi = \psi_0, \lambda \in \Lambda \quad \text{vs.} \quad H_1 : \psi \neq \psi_0, \lambda \in \Lambda$$

the likelihood ratio test statistic is:

$$LR(\mathbf{x}) = \frac{\sup_{\psi, \lambda} L_{\mathbf{X}}(\psi, \lambda; \mathbf{x})}{\sup_{\lambda} L_{\mathbf{X}}(\lambda; \psi_0, \mathbf{x})}.$$

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from an exponential distribution, with probability density function:

$$f_{\mathbf{X}}(x; \mu) = \frac{1}{\mu} \exp\left(-\frac{x}{\mu}\right) \quad \text{for } x \geq 0$$

and 0 otherwise, where $\mu > 0$ is a parameter.

- i. Obtain the test statistic for the likelihood ratio test for testing:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

(7 marks)

- ii. Use the likelihood ratio test statistic to provide the rejection region of an asymptotic test at the $100\alpha\%$ significance level.

(5 marks)

Reading for this question

Section 5.6 of the subject guide presents likelihood ratio tests.

Approaching the question

- i. Due to independence, the joint probability density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \mu) = \prod_{i=1}^n \frac{1}{\mu} \exp\left(-\frac{x_i}{\mu}\right) = \mu^{-n} \exp\left(-\frac{1}{\mu} \sum_{i=1}^n x_i\right)$$

and the log-likelihood function for the observed sample is:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = -n \log(\mu) - \frac{1}{\mu} \sum_{i=1}^n x_i.$$

The score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{X}) = -\frac{n}{\mu} + \frac{1}{\mu^2} \sum_{i=1}^n X_i.$$

The score function is equal to 0 only at $\mu = \bar{X}$. The second derivative of the log-likelihood function is:

$$\frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{n}{\mu^2} - \frac{2n\bar{X}}{\mu^3}$$

which is negative at $\mu = \bar{X}$. Hence we conclude that the likelihood function is maximised at $\bar{X}$.

The likelihood ratio statistic, $LR(\mathbf{X})$ is:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\hat{\mu}; \mathbf{x})}{L_{\mathbf{X}}(\mu_0; \mathbf{x})} = \frac{\hat{\mu}^{-n} \exp\left(-\sum_{i=1}^n X_i / \hat{\mu}\right)}{\hat{\mu}_0^{-n} \exp\left(-\sum_{i=1}^n X_i / \mu_0\right)} = \frac{\bar{X}^{-n} \exp(-n)}{\mu_0^{-n} \exp(-n\bar{X} / \mu_0)}.$$

- ii. Here we should use the asymptotic distribution of the likelihood ratio test. Under H_0 we have:

$$2 \log LR(\mathbf{X}) = 2n(\log(\mu_0) - \log(\bar{X}) - \left(1 - \frac{\bar{X}}{\mu_0}\right)) \sim \chi_1^2$$

approximately. We will reject H_0 for large values of $LR(\mathbf{X})$ which is equivalent to rejecting for large values of $2 \log LR(\mathbf{X})$, since $2 \log(\cdot)$ is an increasing function.

The fact that we now know the distribution of $2 \log LR(\mathbf{X})$ enables us to identify the rejection region of the test. The test should reject H_0 if:

$$2 \log LR(\mathbf{x}) > \chi_{\alpha, 1}^2$$

where $\chi_{\alpha, 1}^2$ denotes the top 100α th percentile of χ_1^2 .